

Karam Baroud

Software Engineer — Full-Stack, Embedded & ML

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Software engineer with production experience across full-stack web, embedded systems, and applied machine learning. Proven record of independently owning technical initiatives, from cloud-deployed platforms to embedded positioning and sensor-fusion tooling, with a strong focus on performance, debugging, and shipping real systems.

EXPERIENCE

Full-Stack Developer — Foundation of the Energy Collective

Jun 2026 - Present · Calgary (Remote)

- Build production software across multiple applications; hired directly from a sponsored capstone project.
- Designed and shipped authentication-gated features using React, Next.js, PostgreSQL, and Keycloak, including Google Sign-In and a volunteer management platform.
- Built an AI content-review tool on the OpenAI API and enhanced a production C# service importing trade data into PostgreSQL; rapidly learned C# to contribute.

Research & Development Intern — ZeroKey Inc.

May 2024 - Jul 2025 · Calgary

Embedded positioning & neural-network IMU systems on a small, high-autonomy R&D team

- Optimized INS positioning engine performance by ~35–50%, cutting per-iteration runtime while moving from ~7ms to ~2–3ms.
- Built an embedded offline replay tool on MCU hardware that cut dataset processing from several minutes to under 30 seconds with exact 1:1 output replication.
- Trained supervised neural-network IMU positioning models on 24 datasets (30,000+ entries each), significantly reducing error during ultrasonic signal loss.

PROJECTS

Asset Tracker FULL-STACK ▶ View

QR-code inventory platform (200+ devices, 10,000+ PPE units) built as capstone lead; deployed and in active use before completion. Led a 5-person team across frontend, backend, database, authentication, and CI/CD.

React · TypeScript · Node.js · PostgreSQL · Keycloak · Docker · GitHub Actions

Optical Sudoku Solver COMPUTER VISION

End-to-end CV pipeline: grid detection, cell extraction, CNN digit recognition, and a backtracking solver.

Python · OpenCV · PyTorch

Tomato Leaf Disease Detection MACHINE LEARNING

Image-classification system using transfer learning (MobileNetV2); handled severe class imbalance with F1-based evaluation.

Python · PyTorch

Squirrel Showdown GAME DEV ▶ Play

Complete game built in a 2-day jam; defend a treehouse from waves of squirrels.

Godot · GDScript

EDUCATION

University of Calgary

B.Sc. Software Engineering

Expected Apr 2026 · GPA 3.87

Dean's List (2022–2024, 2026) · Jason Lang Scholarship (x3) ·

Alexander Rutherford Scholarship

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript, JavaScript, C#, SQL

Frameworks: React, Next.js, Node.js, Express, PyTorch, OpenCV

Cloud/DevOps: Docker, GitHub Actions, AWS Lambda, Terraform, CI/CD

Data: PostgreSQL, MySQL, DynamoDB, Firebase

Focus: Embedded Systems, Sensor Fusion, Computer Vision, ML